

A disproof of Conjecture 00000001273

Periodic-point asymptotics for mixing $\mathbb{Z}^2$ subshifts of finite type

1 The conjecture

Conjecture 00000001273 reads (English version; the Chinese version in `conjectures/00000001273.md` states the same claim):

Conjecture: For any mixing SFT under a $\mathbb{Z}^2$ action, the count of periodic points has main term $\lambda^{t_1 t_2} \cdot (1 + O(2^{-\min t}))$ as $t_1, t_2 \rightarrow \infty$ (periodic-point asymptotics).

We show that it is false under its literal reading, and also under the weaker reading in which “main term” only means that the ratio $P(t_1, t_2)/\lambda^{t_1 t_2}$ tends to 1. One explicit mixing $\mathbb{Z}^2$ SFT is a counterexample to both readings: for every $\lambda > 0$ and every threshold T there are periods $t_1, t_2 \geq T$ at which the error exceeds both the conjectured bound and $\frac{1}{2}\lambda^{t_1 t_2}$. Every step below is a theorem of the accompanying Lean 4 file; the Lean names are given in brackets.

2 Definitions and the reading

Throughout, $A_n = \{0, \dots, n-1\}$ is a finite alphabet and $A_n^{\mathbb{Z}^2}$ is the full shift.

Definition 1 (Shift action; `Config`, `shift`). For $v \in \mathbb{Z}^2$ and $x \in A_n^{\mathbb{Z}^2}$ put $(\sigma^v x)(u) = x(u+v)$. Then $\sigma^0 = \text{id}$ and $\sigma^{v+w} = \sigma^v \sigma^w$, so this is the $\mathbb{Z}^2$ shift action. It is the shift map of the Wikipedia article “Shift space” (https://en.wikipedia.org/wiki/Shift_space), which defines $\sigma^g((x_i)_i) = (x_{gi})_i$ for a group $\mathbb{G}$, here $\mathbb{G} = \mathbb{Z}^2$ written additively.

Definition 2 (Patterns and SFTs; `Pattern`, `Occurs`, `IsSFT`). A *pattern* p is a finite set $F_p \subset \mathbb{Z}^2$ (the window) together with symbols $p(u) \in A_n$ for $u \in F_p$. A configuration x *shows* p at v if $x(v+u) = p(u)$ for all $u \in F_p$. A set $X \subseteq A_n^{\mathbb{Z}^2}$ is a *subshift of finite type* if there is a finite list $\mathcal{F}$ of patterns with

$$X = \{x : \text{no } p \in \mathcal{F} \text{ is shown by } x \text{ at any } v \in \mathbb{Z}^2\}.$$

This is the standard definition. Wikipedia, “Shift space”: “An equivalent way to define a shift space is to take a set of forbidden patterns”. The resulting X_F “is the set of all infinite patterns that do not contain any forbidden finite pattern of” F . The same article continues: “which leads to classical definition of a shift of finite type as those shift spaces” that equal X_F for a finite F . Wikipedia, “Subshift of finite type” (https://en.wikipedia.org/wiki/Subshift_of_finite_type): “subshifts of finite type are shift spaces defined by a finite set of forbidden words”.

Definition 3 (Mixing; `supNorm`, `IsMixing`). Write $\|v\|_\infty = \max(|v_1|, |v_2|)$. A set X is *mixing* if for any two patterns p, q that occur in X (some point of X shows p at 0, some point shows q at 0) there is N such that for every $v \in \mathbb{Z}^2$ with $\|v\|_\infty \geq N$ some $z \in X$ shows p at 0 and q at v .

This is topological mixing of the $\mathbb{Z}^2$ action, written with cylinder sets. Wikipedia, “Mixing (mathematics)” ([https://en.wikipedia.org/wiki/Mixing_\(mathematics\)](https://en.wikipedia.org/wiki/Mixing_(mathematics))): “A system is said to be topologically mixing if, given open sets” A and B , there is N with $f^n(A) \cap B \neq \emptyset$ for all $n > N$. For a non-discrete parameter g (a flow φ_g) the article states the condition “with the requirement that a non-empty intersection hold for all” $\|g\| > N$. We use the same form with $g = v \in \mathbb{Z}^2$ and the sup norm. The cylinders $[p] = \{x : x \text{ shows } p \text{ at } 0\}$ form a basis of the product topology. The point z above lies in $[p] \cap \sigma^{-v}[q] \cap X$, because $\sigma^v z$ shows q at 0 exactly when z shows q at v .

Definition 4 (Periodic points; `perPts`, `perCount`). For $t_1, t_2 \geq 1$ let $\text{Per}_X(t_1, t_2) = \{x \in X : \sigma^{(t_1, 0)}x = x, \sigma^{(0, t_2)}x = x\}$ (the points whose period lattice contains $t_1\mathbb{Z} \times t_2\mathbb{Z}$), and $P_X(t_1, t_2) = \#\text{Per}_X(t_1, t_2)$ (Lean: `Set.ncard`).

Lemma 5 (`perPts_finite`). For every X and all $t_1, t_2 \geq 1$ the set $\text{Per}_X(t_1, t_2)$ is finite, so $P_X(t_1, t_2)$ is the cardinality of a finite set.

Proof. A point $x \in \text{Per}_X(t_1, t_2)$ satisfies $x(a, b) = x(a \bmod t_1, b \bmod t_2)$ (`perPts_apply_emod`, from $x(u + k(t_1, 0)) = x(u)$ and $x(u + k(0, t_2)) = x(u)$ for $k \in \mathbb{Z}$). Hence x is determined by its restriction to $(\mathbb{Z}/t_1) \times (\mathbb{Z}/t_2)$, and the restriction map is injective into the finite set $A_n^{(\mathbb{Z}/t_1) \times (\mathbb{Z}/t_2)}$. $\square$

The two readings. The *literal reading* (`HasPeriodicAsymptotics`, `Conjecture`) is: for every n and every nonempty mixing SFT $X \subseteq A_n^{\mathbb{Z}^2}$ there are $\lambda > 0$, $C \in \mathbb{R}$ and $T \in \mathbb{N}$ such that

$$|P_X(t_1, t_2) - \lambda^{t_1 t_2}| \leq C \cdot 2^{-\min(t_1, t_2)} \cdot \lambda^{t_1 t_2} \quad \text{for all } t_1, t_2 \geq \max(T, 1). \quad (1)$$

This is $P_X = \lambda^{t_1 t_2}(1 + E)$ with $|E| \leq C 2^{-\min t}$, i.e. the main term $\lambda^{t_1 t_2}$ and the hedge $1 + O(2^{-\min t})$ as $t_1, t_2 \rightarrow \infty$. Both λ and the O -constant may depend on X . The *main-term reading* (`HasMainTerm`, `ConjectureMainTerm`) keeps only the main term: there is $\lambda > 0$ such that for every $\varepsilon > 0$ there is T with $|P_X(t_1, t_2) - \lambda^{t_1 t_2}| \leq \varepsilon \lambda^{t_1 t_2}$ for all $t_1, t_2 \geq \max(T, 1)$, i.e. $P_X / \lambda^{t_1 t_2} \rightarrow 1$. The literal reading implies the main-term reading. We refute both directly.

Conventions. Periods are at least 1. For $t = 0$ the set Per_X can be infinite; since T is existential and we also require $t_i \geq 1$, nothing depends on $t = 0$. Requiring $X \neq \emptyset$ only weakens the conjecture: the empty SFT, which has $P = 0$, is excluded, so it cannot serve as a trivial counterexample.

3 The counterexample

Definition 6 (`X3`). $X_3 = \{x \in \{0, 1, 2\}^{\mathbb{Z}^2} : x(i, j) \neq x(i + 1, j) \text{ for all } i, j \in \mathbb{Z}\}$: each row is a proper 3-colouring of $\mathbb{Z}$, and there is no vertical constraint.

Proposition 7 (`X3_nonempty`, `X3_isSFT`). X_3 is nonempty (take $x(i, j) = 0$ for even i and 1 for odd i). It is the SFT whose forbidden patterns are the three horizontal dominoes aa , $a \in \{0, 1, 2\}$, on the window $\{(0, 0), (1, 0)\}$ (*domino*).

Proof. x shows the domino aa at (i, j) iff $x(i, j) = a = x(i + 1, j)$. So no domino occurs iff $x(i, j) \neq x(i + 1, j)$ for all i, j . $\square$

Proposition 8 (`X3_isMixing`). X_3 is mixing.

Proof. Gluing (`glue`, `glue_mem`, `glue_eq_left`, `glue_eq_right`). Let $x, y \in X_3$ and $G \subseteq \mathbb{Z}^2$. Call (i, j) a y -site if $(i, j) \in G$, or if both $(i-1, j) \in G$ and $(i+2, j) \in G$ (`yZone`). Call it a *buffer site* if it is not a y -site but $(i-1, j)$ or $(i+1, j)$ is one (`buffer`). Let $g_0 = y$ on y -sites and $g_0 = x$ elsewhere. Let $g = g_0$ off the buffer sites. At a buffer site (i, j) let $g(i, j)$ be a colour different from both $g_0(i-1, j)$ and $g_0(i+1, j)$ (`third`; it exists because there are three colours).

Two horizontally adjacent sites are never both buffer sites. Suppose $(i, j), (i+1, j)$ both were. Then $(i-1, j)$ and $(i+2, j)$ are y -sites, which forces $(i-1, j) \in G$ and $(i+2, j) \in G$. That makes (i, j) a y -site, a contradiction. Now take a horizontal pair $(i, j), (i+1, j)$:

- If one of them is a buffer site, it differs from the other by the choice of g .
- If neither is a buffer site, they are both y -sites or both not y -sites: a non- y -site next to a y -site is a buffer site. So g agrees there with y or with x , and these differ on the pair.

Hence $g \in X_3$. If (i, j) has sup-distance ≥ 2 from G , then $(i, j), (i \pm 1, j) \notin G$. So (i, j) is neither a y -site nor a buffer site, and $g(i, j) = x(i, j)$. On G we have $g = y$.

Mixing. Let $x \in X_3$ show p at 0 and $y \in X_3$ show q at 0. Let M bound $\|\cdot\|_\infty$ on $F_p \cup F_q$, and let $N = 2M + 2$. For $\|v\|_\infty \geq N$ put $y' = \sigma^{-v}y$, which lies in X_3 and shows q at v . Put $G = v + F_q$. For $a \in F_p$ and $b \in F_q$ we have $\|a - (v + b)\|_\infty \geq N - 2M = 2$. The glued point $z = g$ (built from x, y' and G) lies in X_3 , agrees with x on F_p , and agrees with y' on $v + F_q$. So z shows p at 0 and q at v . $\square$

3.1 Counting periodic points

Let $C(t)$ be the set of proper 3-colourings of the cycle $\mathbb{Z}/t$, i.e. maps $r : \mathbb{Z}/t \rightarrow \{0, 1, 2\}$ with $r(i) \neq r(i+1)$ (`CycCol`), and let $c(t) = \#C(t)$ (`cycCount`).

Lemma 9 (`perEquiv`, `perCount_X3`). *For $t_1, t_2 \geq 1$, $\text{Per}_{X_3}(t_1, t_2)$ is in bijection with the maps $\mathbb{Z}/t_2 \rightarrow C(t_1)$. Hence $P_{X_3}(t_1, t_2) = c(t_1)^{t_2}$.*

Proof. Send x to $b \mapsto (a \mapsto x(a, b))$, with $a \in \mathbb{Z}/t_1$ and $b \in \mathbb{Z}/t_2$ represented by $0 \leq a < t_1$ and $0 \leq b < t_2$. By periodicity $x(a+1, b) = x((a+1) \bmod t_1, b)$, so each row is a proper colouring of the cycle. The inverse sends R to $x(i, j) = R(j \bmod t_2)(i \bmod t_1)$. This point lies in X_3 because $R(\cdot)$ is proper, and it is fixed by $\sigma^{(t_1, 0)}$ and $\sigma^{(0, t_2)}$. The two maps are mutually inverse by Lemma 5's reduction mod (t_1, t_2) . $\square$

Lemma 10 (`transfer matrix`; `pathCount`, `pathCount_succ`, `pathCount_formula`). *Let $N_m(a, b)$ be the number of $r : \{0, \dots, m\} \rightarrow \{0, 1, 2\}$ with $r(0) = a$, $r(m) = b$ and $r(k) \neq r(k+1)$ for $k < m$ (the (a, b) entry of $(J - I)^m$). Then $N_{m+1}(a, b) = \sum_{d \neq a} N_m(d, b)$, and*

$$3N_m(a, b) = 2^m + 2(-1)^m \quad (a = b), \quad 3N_m(a, b) = 2^m - (-1)^m \quad (a \neq b).$$

Proof. Splitting off the first vertex, $r = (a, s)$, gives $r(0) = a$, $r(m+1) = s(m)$, and r is proper iff $a \neq s(0)$ and s is proper. This is the recursion. The closed form holds for $m = 0$ ($N_0(a, b) = [a = b]$). The induction step is $2(2^m - (-1)^m) = 2^{m+1} + 2(-1)^{m+1}$ for $a = b$, and $(2^m + 2(-1)^m) + (2^m - (-1)^m) = 2^{m+1} - (-1)^{m+1}$ for $a \neq b$. $\square$

Proposition 11 (`cycCount_eq_sum`, `cycCount_formula`, `perCount_X3_formula`). *For $t \geq 1$, $c(t) = 2^t + 2(-1)^t$. Hence for $t_1, t_2 \geq 1$*

$$P_{X_3}(t_1, t_2) = (2^{t_1} + 2(-1)^{t_1})^{t_2}.$$

Proof. Write $t = m + 1$ and identify $\mathbb{Z}/(m + 1)$ with $\{0, \dots, m\}$. Then r is a proper cycle colouring iff it is a proper path colouring and $r(m) \neq r(0)$. So $c(m + 1) = \sum_{a \neq b} N_m(a, b)$. There are six ordered pairs $a \neq b$, each with $3N_m(a, b) = 2^m - (-1)^m$. Hence $c(m + 1) = 2(2^m - (-1)^m) = 2^{m+1} + 2(-1)^{m+1}$. $\square$

In particular $P_{X_3}(2k, t_2) = (2^{2k} + 2)^{t_2}$ for $k, t_2 \geq 1$ (`perCount_X3_even`).

4 The refutation

Lemma 12 (`eq_of_pow_comparable`). *Let $a, b > 0$ and $N \in \mathbb{N}$ with $a^t \leq 2b^t$ and $b^t \leq 2a^t$ for all $t \geq N$. Then $a = b$.*

Proof. Suppose $a < b$. Then $b/a > 1$, so $(b/a)^M > 2$ for some M . For $t = \max(M, N)$ we get $b^t = (b/a)^t a^t \geq (b/a)^M a^t > 2a^t$, a contradiction. The case $b < a$ is symmetric. $\square$

Proposition 13 (`core estimate; X3_no_half_bound`). *There are no $\lambda > 0$ and $T \in \mathbb{N}$ with $|P_{X_3}(t_1, t_2) - \lambda^{t_1 t_2}| \leq \frac{1}{2} \lambda^{t_1 t_2}$ for all $t_1, t_2 \geq \max(T, 1)$.*

Proof. Suppose such λ, T exist. Fix $k \geq T + 1$ and put $t_1 = 2k$, $s = 2^{2k} + 2$, $\mu = \lambda^{2k}$. For all $t \geq T + 1$ the hypothesis gives $|s^t - \mu^t| \leq \frac{1}{2} \mu^t$, so $s^t \leq 2\mu^t$ and $\mu^t \leq 2s^t$. The lemma gives $\mu = s$:

$$\lambda^{2k} = 2^{2k} + 2 \quad \text{for every } k \geq T + 1.$$

Take $k = T + 1, T + 2, T + 3$ and put $A = 2^{2(T+1)} > 0$, $L = \lambda^2$. Then $\lambda^{2(T+1)} = A + 2$, $(A + 2)L = 4A + 2$ and $(A + 2)L^2 = 16A + 2$. Therefore

$$18A = (4A + 2 + (A + 2)L)((A + 2)L - (4A + 2)) - (A + 2)((A + 2)L^2 - (16A + 2)) = 0,$$

contradicting $A > 0$. $\square$

Theorem 14 (`not_hasMainTerm_X3, not_hasPeriodicAsymptotics_X3`). *X_3 satisfies neither the main-term reading nor the literal reading (1).*

Proof. Main term: take $\varepsilon = \frac{1}{2}$ and apply Proposition 13. Literal: given $\lambda > 0$, C and T , choose N with $2^{-N} < 1/(2(|C| + 1))$. For $t_1, t_2 \geq \max(T, N)$ we get $C 2^{-\min(t_1, t_2)} \leq (|C| + 1)2^{-N} \leq \frac{1}{2}$. Then (1) implies the bound of Proposition 13 with threshold $\max(T, N)$, which is impossible. $\square$

Equivalently, along an infinite family of periods (`X3_violates_periodicAsymptotics, X3_violates_mainTerm`): for every $\lambda > 0$, every C and every T there are $t_1, t_2 \geq \max(T, 1)$ with $|P_{X_3}(t_1, t_2) - \lambda^{t_1 t_2}| > C 2^{-\min(t_1, t_2)} \lambda^{t_1 t_2}$. For every $\lambda > 0$ and T there are $t_1, t_2 \geq \max(T, 1)$ with $|P_{X_3}(t_1, t_2)/\lambda^{t_1 t_2} - 1| > \frac{1}{2}$.

Theorem 15 (`conjecture_00000001273_false, conjecture_00000001273_mainTerm_false, conjecture_00000001273_disproved`). *Both readings of Conjecture 00000001273 are false. X_3 is a nonempty mixing $\mathbb{Z}^2$ SFT over the alphabet $\{0, 1, 2\}$, with finitely many periodic points of each period. It admits no $\lambda > 0$ satisfying (1), nor any $\lambda > 0$ with $P_{X_3}(t_1, t_2)/\lambda^{t_1 t_2} \rightarrow 1$.*

Remark 16. The obstruction is that the “main term” cannot be of the form $\lambda^{t_1 t_2}$. For fixed t_1 the count grows like $c(t_1)^{t_2}$ in t_2 , and $c(t_1)^{1/t_1} = (2^{t_1} + 2(-1)^{t_1})^{1/t_1}$ is not constant in t_1 . The entropy of X_3 is $\log 2$, but $\lambda = 2$ fails like every other λ : for even t_1 , $P/2^{t_1 t_2} = (1 + 2^{1-t_1})^{t_2} \rightarrow \infty$ as $t_2 \rightarrow \infty$.

5 Lean 4 formalization

The directory `lean/` is a Lean 4 project (Lean v4.33.1, Mathlib v4.33.1; package `conjecture1273`, library `Conjecture1273`). `Conjecture1273.lean` imports the single source file `Conjecture1273/Basic.lean`, namespace `Conjecture1273`.

Definitions.

- `Config n := Int x Int -> Fin n`; `shift v x := fun u => x (u + v)`.
- `Pattern n: support : Finset (Int x Int), val : Int x Int -> Fin n`; `Occurs x p v := forall u in p.support, x (v + u) = p.val u`.
- `IsSFT X := exists forbidden : List (Pattern n), X = {x | forall p in forbidden, forall v, not Occurs x p v}`.
- `supNorm v := max |v.1| |v.2| (as natAbs)`; `IsMixing X`: for all patterns `p q` occurring in `X` at 0 there is `N : Nat` such that every `v` with `N <= supNorm v` has some `z in X` with `Occurs z p 0` and `Occurs z q v`.
- `perPts X t1 t2 := {x | x in X /\ shift (t1,0) x = x /\ shift (0,t2) x = x}`; `perCount X t1 t2 := (perPts X t1 t2).ncard`.
- `HasPeriodicAsymptotics X, HasMainTerm X, Conjecture, ConjectureMainTerm` as in Section 2, with `lam : Real, 0 < lam, 2-min` written `(1/2 : Real) ^ min t1 t2`, and the quantifier `forall t1 t2, 1 <= t1 -> 1 <= t2 -> T <= t1 -> T <= t2 -> ....`

Main theorem. `conjecture_00000001273_false` states the literal conjecture in full and negates it:

$$\neg \forall n (X \subseteq \text{Config } n), X \neq \emptyset \rightarrow \text{IsSFT } X \rightarrow \text{IsMixing } X \rightarrow \exists \lambda > 0, \exists C, \exists T, \\ \forall t_1 t_2, 1 \leq t_1 \rightarrow 1 \leq t_2 \rightarrow T \leq t_1 \rightarrow T \leq t_2 \rightarrow \\ |\text{perCount } X t_1 t_2 - \lambda^{t_1 t_2}| \leq C \cdot (1/2)^{\min(t_1, t_2)} \cdot \lambda^{t_1 t_2}.$$

Its proof is `not_hasPeriodicAsymptotics_X3 (h 3 X3 X3_nonempty X3_isSFT X3_isMixing). conjecture_00000001273_mainTerm_false` does the same for the main-term reading, and `conjecture_00000001273_disproved : not Conjecture /\ not ConjectureMainTerm` combines the two.

Other theorems. `perPts_finite` (Lemma 5); `X3_nonempty`, `X3_isSFT`, `X3_isMixing` (with `glue_mem`, `glue_eq_left`, `glue_eq_right`); `perEquiv`, `perCount_X3`, `pathCount_succ`, `pathCount_formula`, `cycCount_eq_sum`, `cycCount_formula`, `perCount_X3_formula`, `perCount_X3_even`; `eq_of_pow_comparable`, `X3_no_half_bound`, `not_hasMainTerm_X3`, `not_hasPeriodicAsymptotics_X3`, `X3_violates_periodicAsymptotics`, `X3_violates_mainTerm`. `X3_counterexample` is the conjunction: `nonempty`, `SFT`, `mixing`, finitely many periodic points of each period, the count formula, and the failure of both readings.

Axiom audit. There is no `sorry`, no `native_decide` and no added axiom. `#print axioms` for `conjecture_00000001273_false`, `conjecture_00000001273_mainTerm_false`, `conjecture_00000001273_disproved` and `X3_counterexample` reports only `propext`, `Classical.choice` and `Quot.sound`. The file also elaborates without warnings under `-DwarningAsError=true`. To check:

```
cd lean
lake exe cache get
lake build
```